Optimizing care and reducing cost: the impact of health-system specialty pharmacists in idiopathic pulmonary fibrosis management

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Background

- Patients with idiopathic pulmonary fibrosis (IPF) are often hospitalized secondary to respiratory worsening or acute exacerbation.[1]
- The medical costs associated with IPF can lead to substantial economic burden, with annual estimates of around $110 million in the U.S.[1]
- The involvement of health-system specialty pharmacies (HSSPs) within multidisciplinary teams has demonstrated cost avoidance in other specialties[2]

Objectives

To evaluate potential costs avoided due to HSSP pharmacist interventions performed for patients diagnosed with IPF

Methods

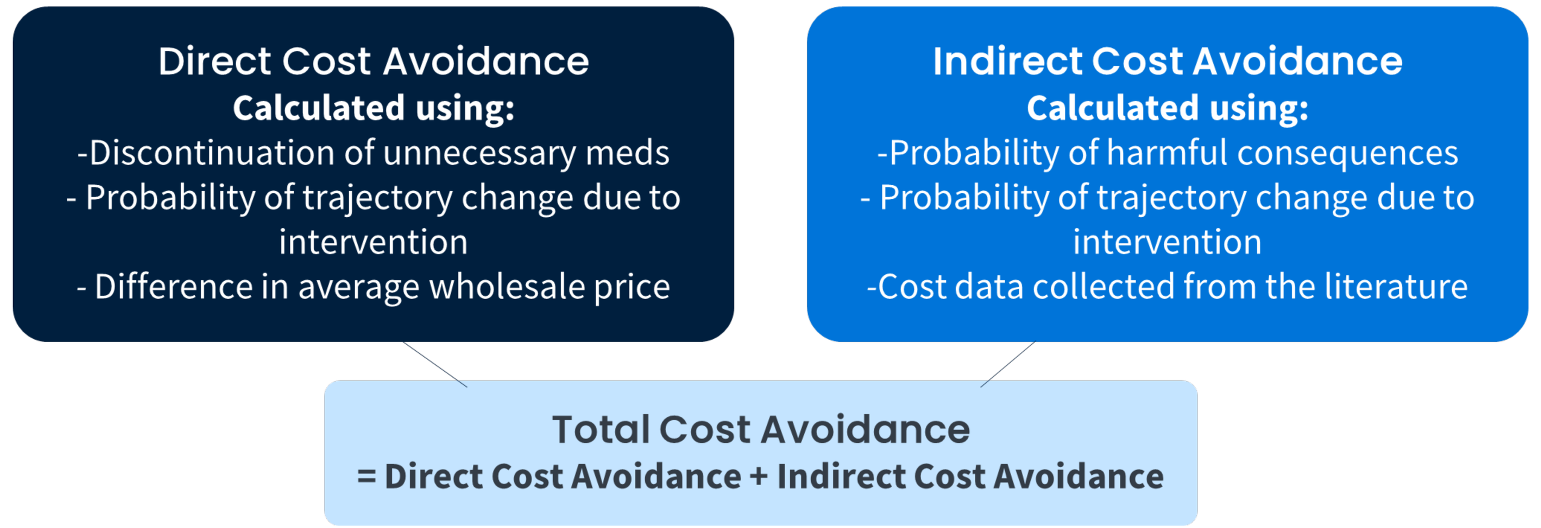
Study Design

This retrospective, observational cohort study was conducted at twelve client health systems of CPS Solutions, LLC (CPS) from April 2023 to April 2024.

Inclusion Criteria	Exclusion Criteria
• Interventions completed for patients who were diagnosed with IPF • Interventions for an IPF-related specialty therapy	• Incomplete documentation • Duplicate Interventions • Interventions with insufficient documentation to accurately estimate cost avoidance

Outcomes

Cost avoidance was calculated using the method recommended by Patanwala et al.[3]



Results

Table 1: Patient Demographics

Characteristics	N=92
Age, years (Mean ± SD)	73 ± 9.4
Gender	
Female (n (%))	45 (49)
Male (n (%))	47 (51)

Figure 2: Evaluated Therapies (N=139)

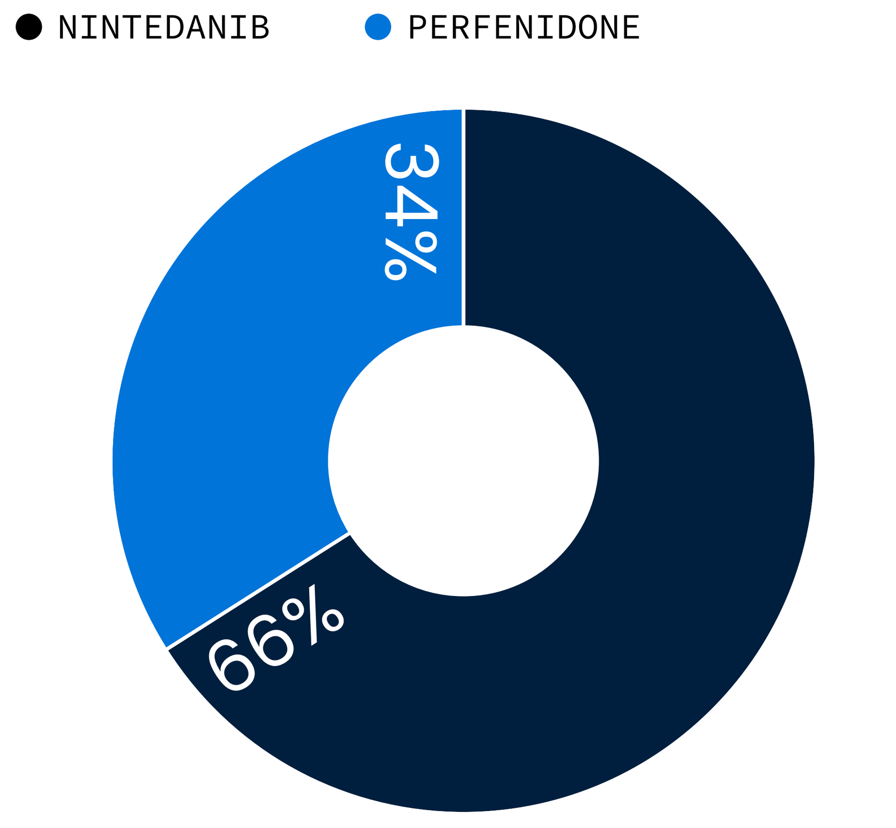
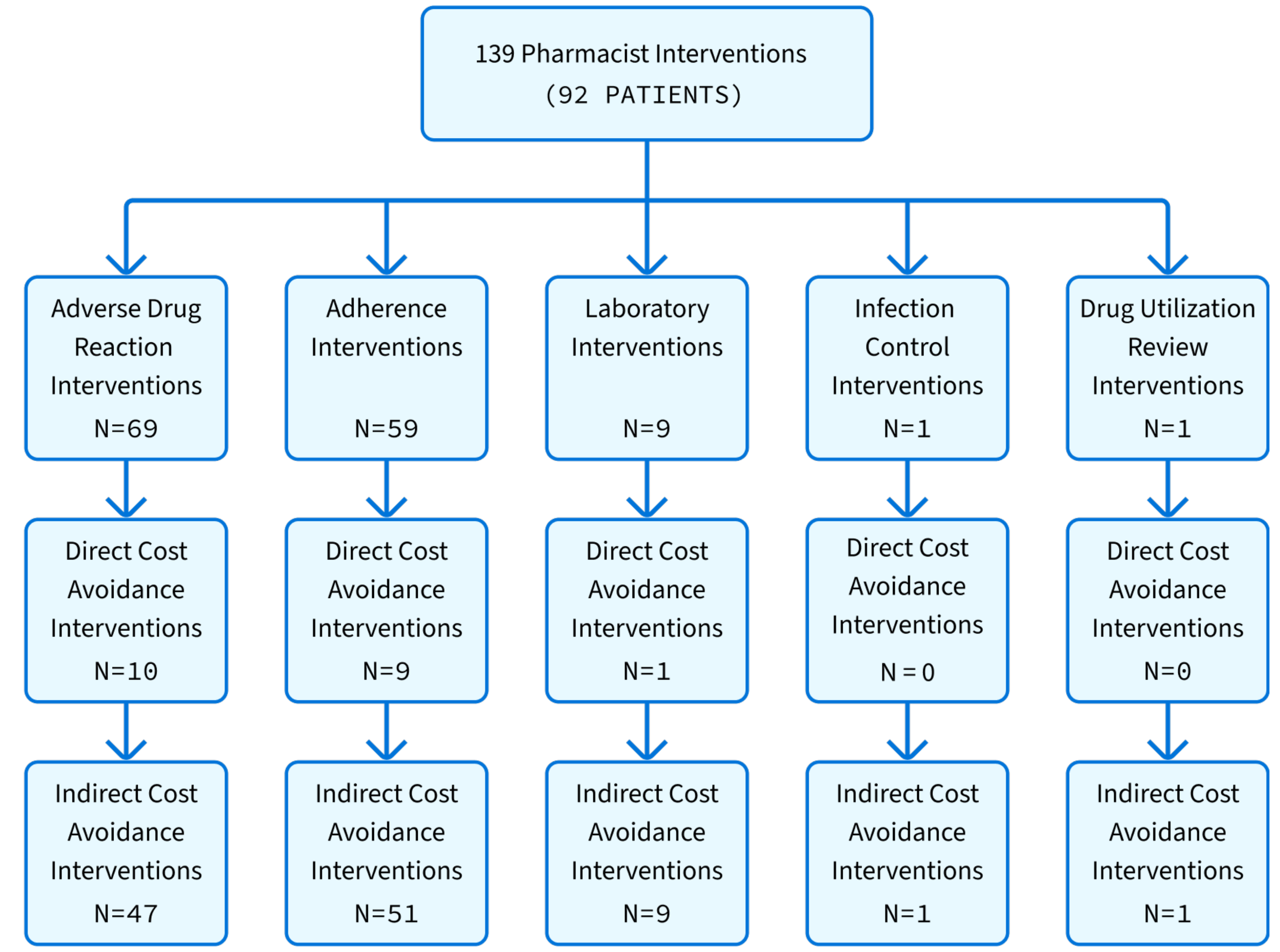


Figure 3: HSSP Intervention Types



Results Cont.

Table 2: Cost Avoidance Associated with HSSP Interventions

Cost Avoidance	Overall Cost	Lower Limit	Upper Limit
Direct Cost Avoidance	$191,542	$45,649	$215,151
Indirect Cost Avoidance	$21,879	$1,455	$24,054
Total Cost Avoidance	**$213,421**	**$47,105**	**$239,211**

Discussion & Conclusion

- HSSP teams who provide care for patients with IPF perform clinically meaningful interventions, and this study demonstrates the impact of these services by quantifying the potential costs avoided as a result.
- Over the course of one year, HSSP interventions at IPF outpatient clinics resulted in an estimated cost avoidance of approximately $213,000.
- One limitation of this study is that pharmacists' labor costs were not accounted for, which may affect the overall estimation of total cost avoidance.
- Future studies should evaluate the financial impact of pharmacist interventions across a larger sample size and different disease states.

References

1 Mooney JJ, Raimundo K, Chang E, Broder MS. Hospital cost and length of stay in idiopathic pulmonary fibrosis. *J Med Econ*. 2017;20(5):518-524. doi:10.1080/13696998.2017.1282864

2 Georgieva D, Markley B, DeClercq J, Choi L, Zuckerman AD. Cost avoidance from health system specialty pharmacist interventions in patients with multiple sclerosis. *J Manag Care Spec Pharm*. 2024;30(4):336-344. doi:10.18553/jmcp.2024.30.4.336

3 Patanwala AE, Narayan SW, Haas CE, Abraham I, Sanders A, Erstad BL. Proposed guidance on cost-avoidance studies in pharmacy practice. *Am J Health Syst Pharm*. 2021;78(17):1559-1567. doi:10.1093/ajhp/zxab211